

Metagenomic assemblies - what's next?

Rui Guan | Sina Beier | Lajos Kalmar

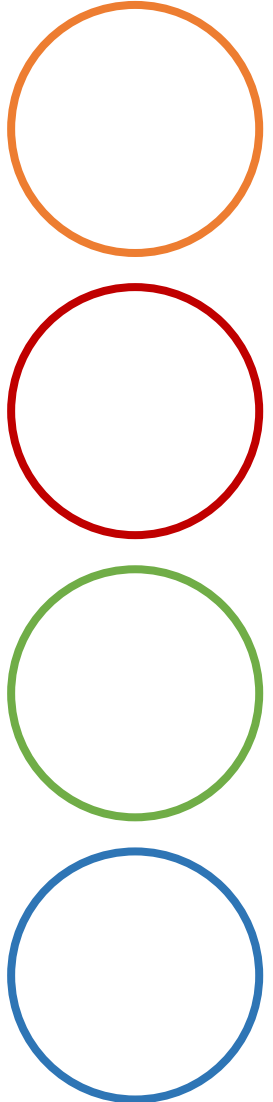
University of Cambridge

2025.11.24

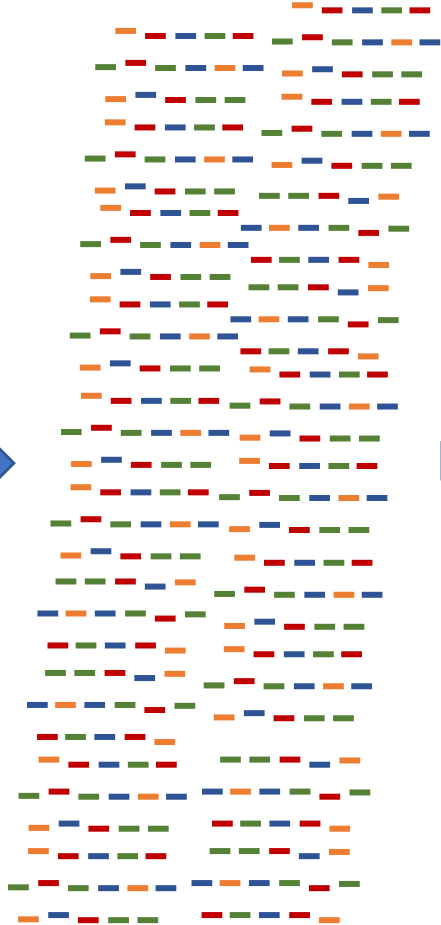
Recap

Binning – finishing the puzzle

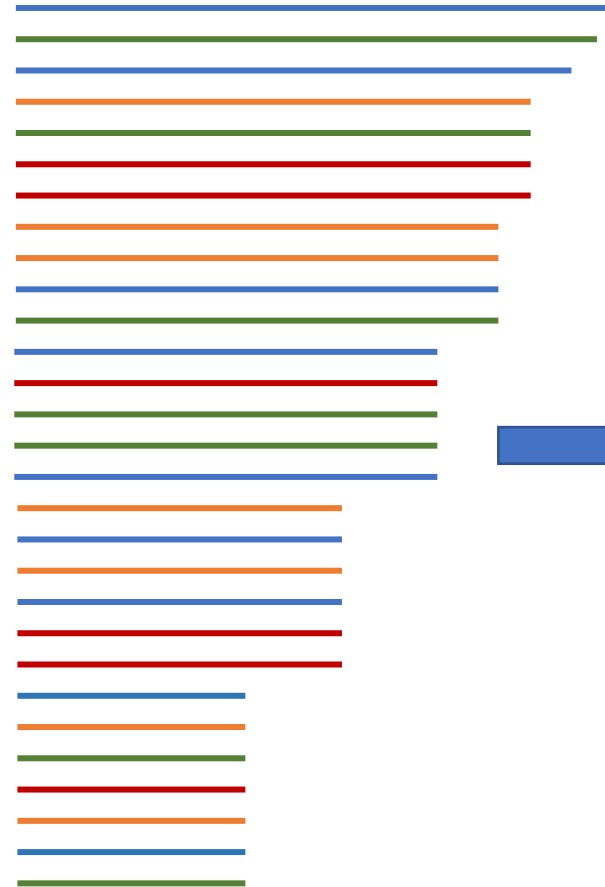
Original genomes



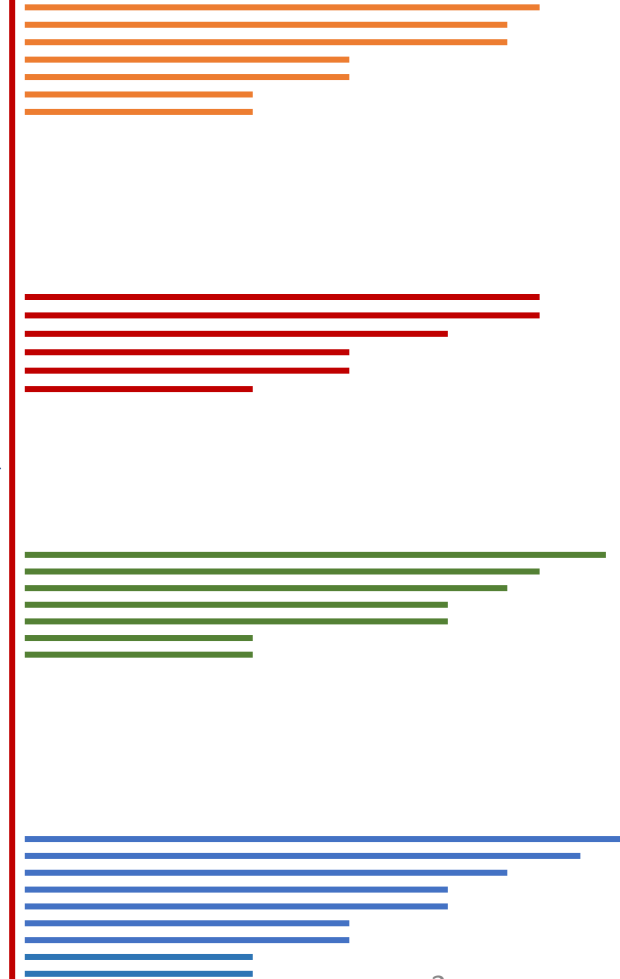
Shotgun sequencing



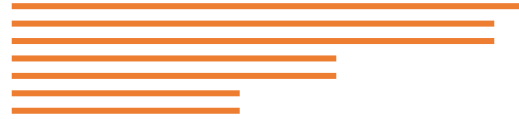
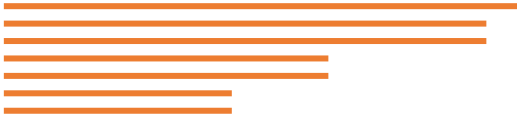
De-novo assembly



Binned final assemblies



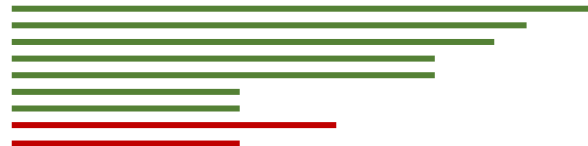
Metagenome assembled genomes (MAGs)



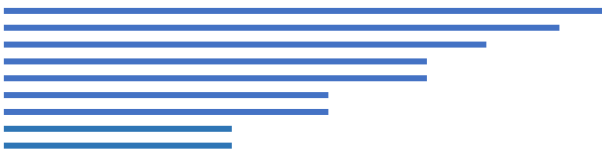
Complete, non-contaminated MAG



Incomplete MAG



Contaminated MAG



Complete, non-contaminated MAG
representing an unknown species

(Meta)genomics post-processing steps

Working with metagenome assembled genomes (MAGs)

- Quality control, quality comparison
- Identification
- Annotation – general and functional
- Comparison to close / distant homologues
- Visualisation
- (Reporting)

Post-assembly QC

Quality control and comparison

- Simple unix commands: `wc`, `cut`, `grep`, ...
- Reference based quality assessment (QUAST, MetaQUAST)
- Single copy gene based assessment (CheckM, GTDB-tk)
- Try to evaluate multiple samples at the same time and compare them to each other
- Consider accepted measures but also build up your own rules

Quality control - simple unix commands

- File size gives a good idea about the amount of nucleotides (genome size) in the MAGs

```
ls -l mag_001
```

```
ls -lh mag_001
```

- Number of contigs in the MAG

```
grep -c ">" mag_001.fasta
```

- The FASTA header of the 30 longest contigs

```
grep ">" mag_001.fasta | sort -gr -t "_" -k 6 | head -n 30
```

- Number of contigs longer than 10k bp

```
grep ">" mag_001.fasta | cut -d "_" -f 4 | awk '$1 > 10000' |  
wc -l
```


Quality control - reference based quality assessment

QUAST/ MetaQUAST

JOURNAL ARTICLE

QUAST: quality assessment tool for genome assemblies FREE

Alexey Gurevich ✉, Vladislav Saveliev, Nikolay Vyahhi, Glenn Tesler [Author Notes](#)

Bioinformatics, Volume 29, Issue 8, April 2013, Pages 1072–1075,

<https://doi.org/10.1093/bioinformatics/btt086>

Published: 19 February 2013 **Article history** ▼

JOURNAL ARTICLE

MetaQUAST: evaluation of metagenome assemblies FREE

Alla Mikheenko, Vladislav Saveliev, Alexey Gurevich ✉ [Author Notes](#)

Bioinformatics, Volume 32, Issue 7, April 2016, Pages 1088–1090,

<https://doi.org/10.1093/bioinformatics/btv697>

Published: 26 November 2015 **Article history** ▼

Quality control - reference based quality assessment

QUAST

PLOS ONE

 OPEN ACCESS  PEER-REVIEWED

RESEARCH ARTICLE

Plantagora: Modeling Whole Genome Sequencing and Assembly of Plant Genomes

Roger Barthelson , Adam J. McFarlin, Steven D. Rounsley, Sarah Young

Published: December 12, 2011 • <https://doi.org/10.1371/journal.pone.0028436>

Quality control - reference based quality assessment

QUAST

Quality metrics:

1) Contig sizes

- No. of contigs, largest contig, total length, Nx, NGx

2) Mis-assemblies and structural variations

3) Genome representation and its functional elements

4) Variations of N50 based on aligned blocks

From QUAST to MetaQUAST

- 1) detect and download reference sequences of unknown species content (limited to bacteria and archaea)
- 2) (in theory) unlimited number of reference genomes
- 3) inter-species misassembly - chimeric contigs
- 4) redesigned visualization

Quality control - reference based quality assessment

QUAST / MetaQUAST

- Aligns the contigs to known NCBI reference genomes and predicts the MAG completeness, contamination
- Works very well for known genomes / communities
- Graphical user interface, reporting, also [web interface](#)
- Risk of bias for less known species, communities

Quality control - single copy gene based assessment using CheckM and GTDB-tk

- General MAG statistics (No. contigs, genome length, N50, GC content, etc.)
- (*almost*) Unbiased measures for genome completeness and contamination
- Checking the presence and copy number of known **essential single copy genes** (N = 104)
- Low amount of single copy genes -> reduced completeness
- Multiple copies of single copy genes -> increased contamination

Quality control - single copy gene based assessment using CheckM and GTDB-tk

- Lineage-specific Workflow - quality estimates with lineage-specific markers
- Taxonomic-specific Workflow - quality estimates with taxonomy-specific markers
- Using Custom Marker Genes - genome quality estimates with custom markers
- Using CPR Marker Set - genome quality estimates with a Candidate phyla radiation (CPR) / Patescibacteria specific marker set (N = 43)



Welcome to GTDB

GENOME TAXONOMY DATABASE

901,341 genomes

Release 11-RS232 (15th April 2026)



Australian Government
Australian Research Council



Quality control - single copy gene based assessment using CheckM and GTDB-tk

GTDB-tk is resource demanding due to the size of database!

Hardware requirements

Domain	Memory	Storage	Time
Archaea	~45 GB	~85 GB	~1 hour / 1,000 genomes @ 64 CPUs
Bacteria	~65GB (410 GB when using -full_tree)	~85 GB	~1 hour / 1,000 genomes @ 64 CPUs

Quality control - CheckM and CheckM2

Method



CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes

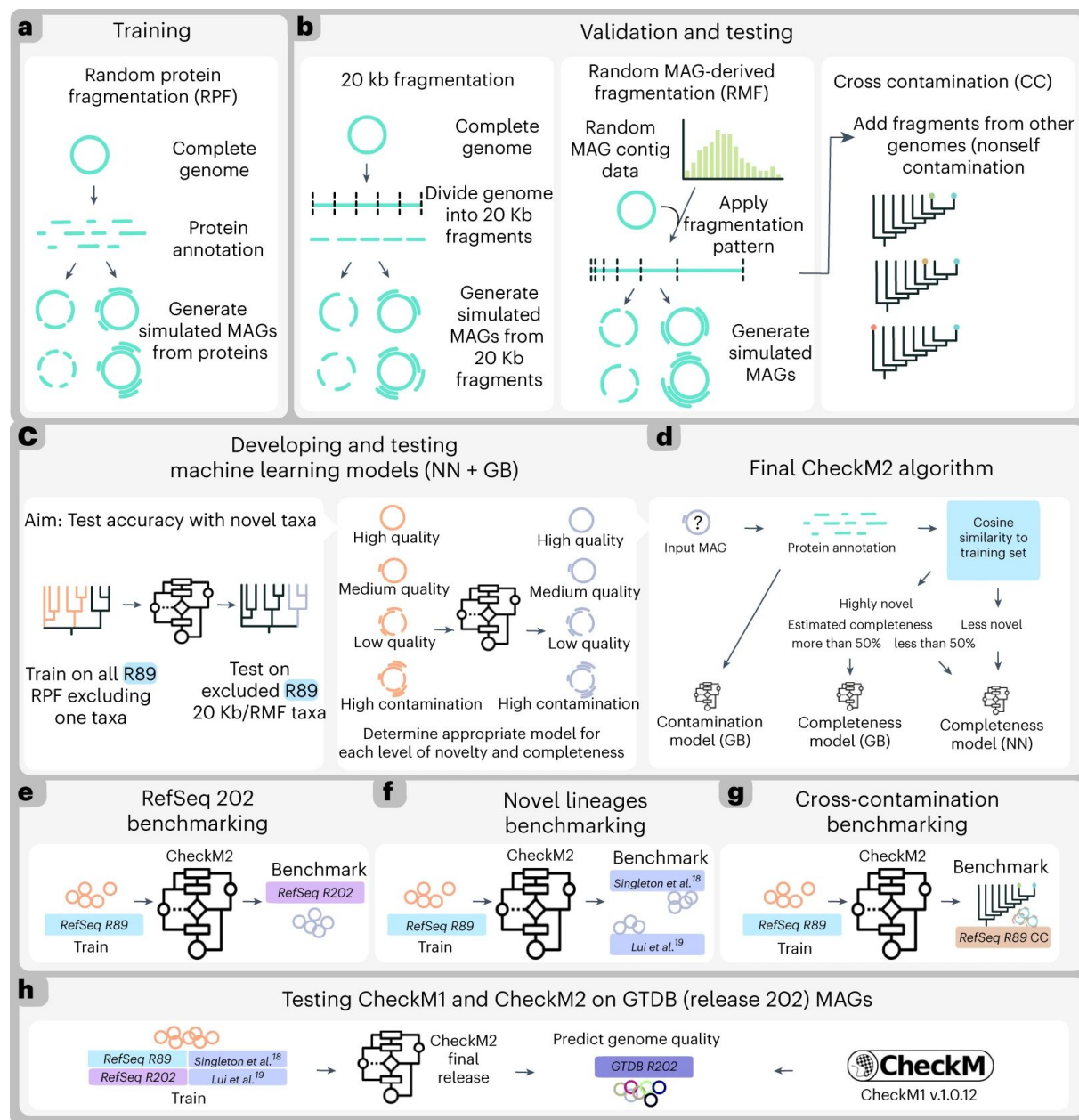
Donovan H. Parks,¹ Michael Imelfort,¹ Connor T. Skennerton,¹ Philip Hugenholtz,^{1,2} and Gene W. Tyson^{1,3}

CheckM2: a rapid, scalable and accurate tool for assessing microbial genome quality using machine learning

[Alex Chklovski](#), [Donovan H. Parks](#), [Ben J. Woodcroft](#) & [Gene W. Tyson](#) 

[Nature Methods](#) **20**, 1203–1212 (2023) | [Cite this article](#)

Overview of CheckM2 development, benchmarking and validation



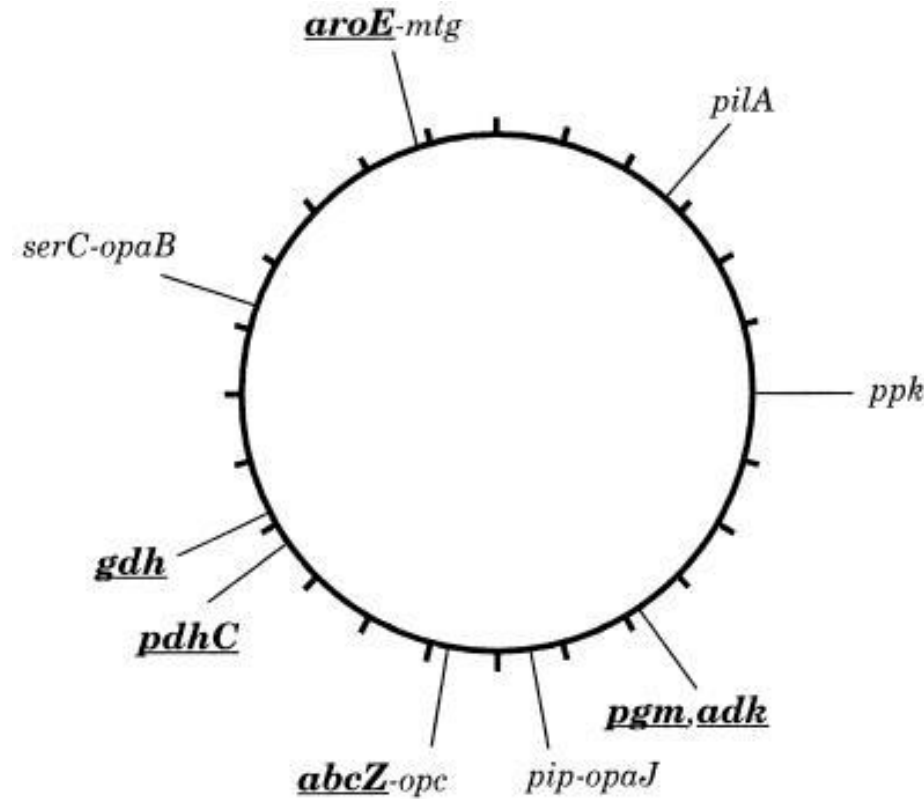
Annotation

Annotation - identification of species and strain

- BLASTn
- *Kmer* based methods (`mash`, `kraken`, `bracken`)
- Multi-Locus Sequence Typing (MLST, `mlst`)
- Species / genus specific identifiers (`legsta` for *Legionella pneumophila*)
- Compare to your own isolates / databank (high coverage `mash`)

Annotation - identification of strain

- Multi-Locus Sequence Typing (MLST, *mlst*)



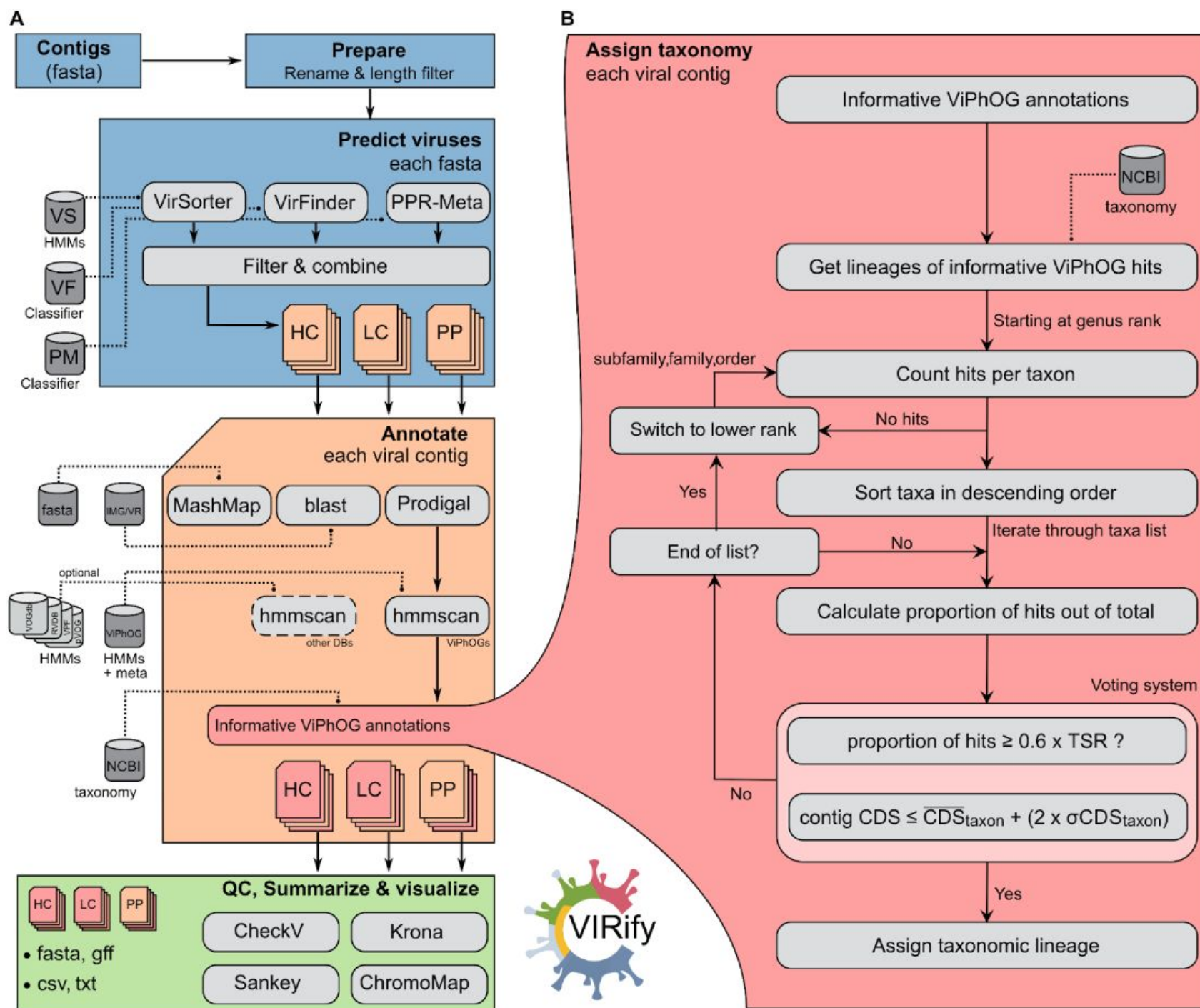
Chromosomal locations of gene fragments.

Annotation - identification of virus

- Extra steps required to capture the sequences - e.g. RNA virus
- High rates of recombination
- No universal marker genes
- Underrepresented databases



Workflow overview of VIRify



Annotation - whole genome annotation

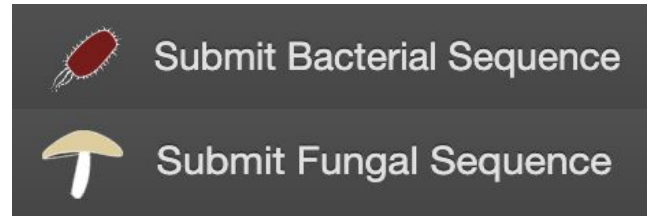
- Finding open reading frames
- Identify gene-like sequences
- Find homologous known genes to identify gene names and functions
- Potentially lot of “hypothetical” or “unknown” named genes for less known species
- *De novo* prokaryotic genome annotation (`prokka`, `bakta`)

Annotation – functional annotation

- Usually working with databases, pay attention to the download location it can take a lot of storage space!
- Antimicrobial resistance gene annotation (ABRicate)
- Plasmid identification (ABRicate, PlasFlow, PlasForest)
- Virulence gene identification (ABRicate)
- Metabolic pathway identification (METABOLIC)

Annotation - secondary metabolite biosynthesis gene clusters

antiSMASH - antibiotics & Secondary Metabolite Analysis Shell



antiSMASH 7.0: new and improved predictions for detection, regulation, chemical structures, and visualisation

Kai Blin, Simon Shaw, Hannah E Augustijn, Zachary L Reitz, Friederike Biermann, Mohammad Alanjary, Artem Fetter, Barbara R Terlouw, William W. Metcalf, Eric JN Helfrich, Gilles P van Wezel, Marnix. H Medema & Tilmann Weber

Nucleic Acids Research (2023) doi: [10.1093/nar/gkad344](https://doi.org/10.1093/nar/gkad344).

antiSMASH: Rapid identification, annotation and analysis of secondary metabolite biosynthesis gene clusters.

Marnix H. Medema, Kai Blin, Peter Cimermancic, Victor de Jager, Piotr Zakrzewski, Michael A. Fischbach, Tilmann Weber, Rainer Breitling & Eriko Takano

Nucleic Acids Research (2011) doi: [10.1093/nar/gkr466](https://doi.org/10.1093/nar/gkr466)

Annotation – Mobile Genetic Elements (MGEs)

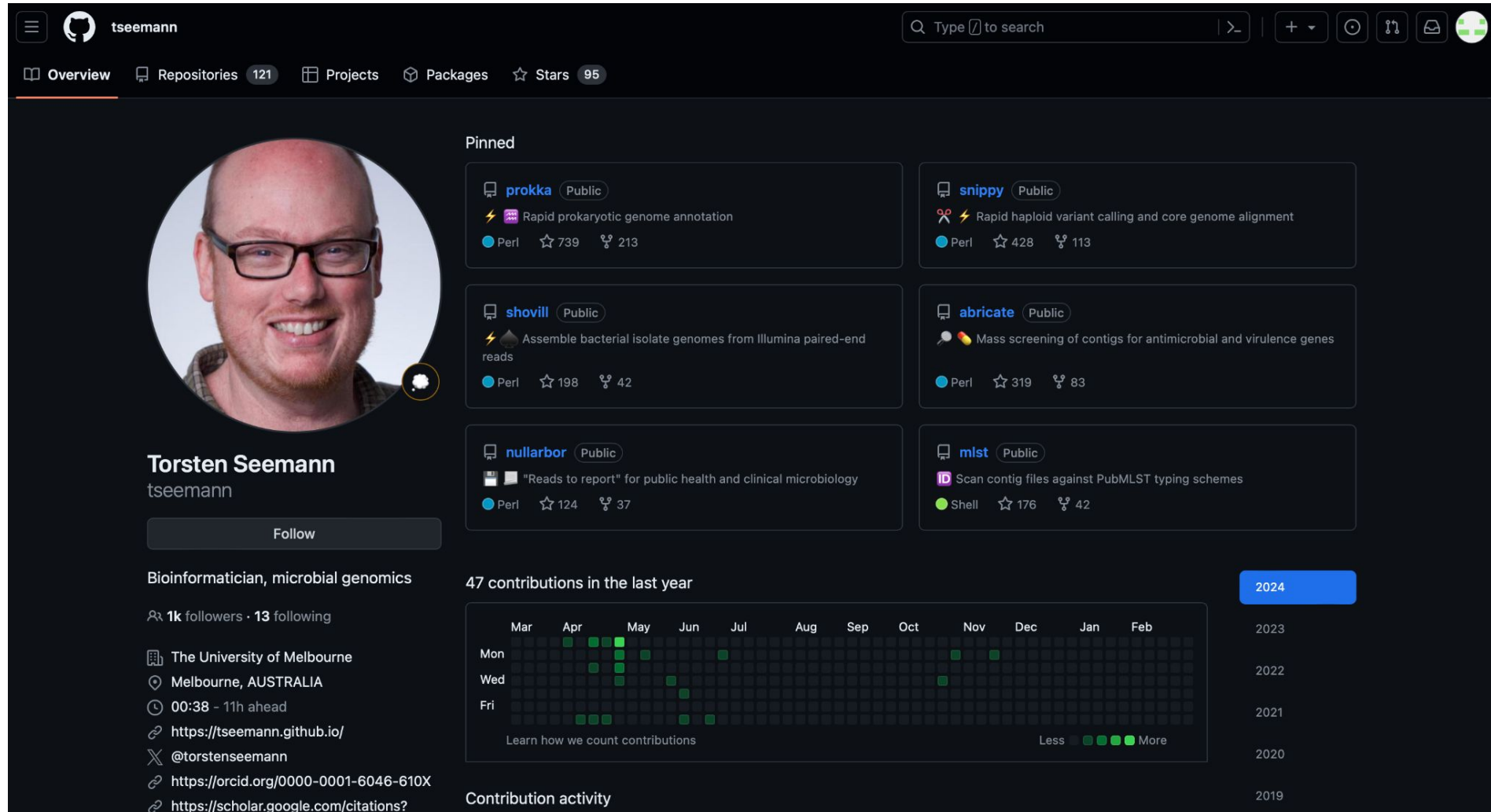
- MGEs -> spread of AntiMicrobial Resistance (AMR) genes
- geNomad
 - Gene content information + deep neural network
 - Identification of plasmids and viruses

Identification of mobile genetic elements with geNomad

[Antonio Pedro Camargo](#) ✉, [Simon Roux](#), [Frederik Schulz](#), [Michal Babinski](#), [Yan Xu](#), [Bin Hu](#),
[Patrick S. G. Chain](#), [Stephen Nayfach](#) & [Nikos C. Kyrpides](#) ✉

[Nature Biotechnology](#) (2023) | [Cite this article](#)

The bioinformatician in classic (meta)genomics



The screenshot shows the GitHub profile of Torsten Seemann. The profile includes a circular profile picture of a man with glasses and a beard. Below the picture is the name "Torsten Seemann" and the username "tseemann". A "Follow" button is visible. The bio states "Bioinformatician, microbial genomics". The profile shows 1k followers and 13 following. The location is "The University of Melbourne, Melbourne, AUSTRALIA". The time zone is "00:38 - 11h ahead". The profile includes links to the GitHub repository, Twitter, ORCID, and Google Scholar. The "Pinned" section displays six repositories: **prokka** (Rapid prokaryotic genome annotation, Perl, 739 stars, 213 forks), **snippy** (Rapid haploid variant calling and core genome alignment, Perl, 428 stars, 113 forks), **shovill** (Assemble bacterial isolate genomes from Illumina paired-end reads, Perl, 198 stars, 42 forks), **abricate** (Mass screening of contigs for antimicrobial and virulence genes, Perl, 319 stars, 83 forks), **nullarbor** ("Reads to report" for public health and clinical microbiology, Perl, 124 stars, 37 forks), and **mlst** (Scan contig files against PubMLST typing schemes, Shell, 176 stars, 42 forks). The "47 contributions in the last year" section shows a calendar grid with green squares indicating contributions. The "Contribution activity" section shows a bar chart of contributions over time.

torsten seemann

Overview Repositories 121 Projects Packages Stars 95

Torsten Seemann
tseemann

Follow

Bioinformatician, microbial genomics

1k followers · 13 following

The University of Melbourne
Melbourne, AUSTRALIA
00:38 - 11h ahead
<https://tseemann.github.io/>
@torstenseemann
<https://orcid.org/0000-0001-6046-610X>
<https://scholar.google.com/citations?>

Pinned

- prokka** Public
⚡️ Rapid prokaryotic genome annotation
Perl 739 stars 213 forks
- snippy** Public
⚡️ Rapid haploid variant calling and core genome alignment
Perl 428 stars 113 forks
- shovill** Public
⚡️ Assemble bacterial isolate genomes from Illumina paired-end reads
Perl 198 stars 42 forks
- abricate** Public
🔍 Mass screening of contigs for antimicrobial and virulence genes
Perl 319 stars 83 forks
- nullarbor** Public
📄 "Reads to report" for public health and clinical microbiology
Perl 124 stars 37 forks
- mlst** Public
🔍 Scan contig files against PubMLST typing schemes
Shell 176 stars 42 forks

47 contributions in the last year

2024

2023

2022

2021

2020

2019

Contribution activity

(meta)Genomic annotation formats

For the computer

(post-processing, comparison, visualisation)

GFF3, tabular formats (.csv, .tsv), json, xml

GFF example of a gene and its graphical representation



chr1	tool	gene	11218	15435	.	+	.	ID=gene1
chr1	tool	mRNA	11218	15435	.	+	.	ID=transcript1;Parent=gene1
chr1	tool	exon	11218	13000	.	+	.	ID=exon1;Parent=transcript1
chr1	tool	exon	13800	14002	.	+	.	ID=exon2;Parent=transcript1
chr1	tool	exon	15000	15360	.	+	.	ID=exon3;Parent=transcript1
chr1	tool	exon	15384	15435	.	+	.	ID=exon4;Parent=transcript1
chr1	tool	UTR5	11218	12000	.	+	.	ID=UTR5a;Parent=transcript1
chr1	tool	CDS	12801	13000	.	+	0	ID=CDS1;Parent=transcript1
chr1	tool	CDS	13800	14002	.	+	0	ID=exon1;Parent=transcript1
chr1	tool	CDS	15000	15234	.	+	0	ID=exon1;Parent=transcript1
chr1	tool	UTR3	15234	15360	.	+	.	ID=UTR3a;Parent=transcript1
chr1	tool	UTR3	15384	15435	.	+	.	ID=UTR3b;Parent=transcript1

For the eye (human readable format)

.gbk (GeneBank format)

```

Rouze,P., Sandberg,G., Segura,M.D., Serizet,C., Tabrett,A.,
Taconnat,L., Thareau,V., Van Hummelen,P., Vercruysse,S.,
Vuylsteke,M., Weingartner,M., Weisbeek,P.J., Wirta,V.,
Wittink,F.R., Zabeau,M. and Small,I.
TITLE      Versatile gene-specific sequence tags for Arabidopsis functional
            genomics: transcript profiling and reverse genetics applications
JOURNAL    Genome Res. 14 (10B), 2176-2189 (2004)
PUBMED     15489341
REFERENCE  2 (bases 1 to 10459)
AUTHORS    Small,I.D. and Hilson,P.
TITLE      pAGRIKOLA sequence
JOURNAL    Unpublished
REFERENCE  3 (bases 1 to 10459)
AUTHORS    Small,I.D. and Hilson,P.
TITLE      Direct Submission
JOURNAL    Submitted (07-MAR-2004) URGV, INRA/CNRS/UEVE, 2, rue Gaston
            Cremieux, CP5708, Evry, Essonne 91057, France
FEATURES   Location/Qualifiers
            source          1..10459
                        /organism="Cloning vector pAGRIKOLA"
                        /mol_type="other DNA"
                        /db_xref="taxon:267887"
                        /lab_host="Escherichia coli"
                        /note="high-throughput plant RNAi cloning vector"
            rep_origin      18..672
                        /note="colE1"
            gene            complement(820..1635)
                        /gene="nptI"
            CDS             complement(820..1635)
                        /gene="nptI"

```

Comparative metagenomics

Comparison to other genomes

- Find homologues by `BLAST`
- Use your own archive collection
- Compare multiple samples within one project

- Whole genome alignments are resource and time hungry
- Gene level comparison -> pan-genome (`roary`, `panaroo`)
- Core genome, accessory genome

MAG catalogues - MGnify

- Biome-specific, clustered, annotated collections of draft genomes
- Submit, analyse, discover and compare microbiome data
- MGnify
 - Average Nucleotide Identity (ANI) = 95% “species”
 - Representative of a cluster
 - Species representatives are assigned the unique MGYG ID



Human
(189359)



Digestive
system
(109405)



Aquatic
(51107)



Marine
(37646)



Digestive
system
(35301)



Plants
(28584)



Soil
(25901)



Skin
(11295)



Wastewater
(4346)



Food
production
(2919)

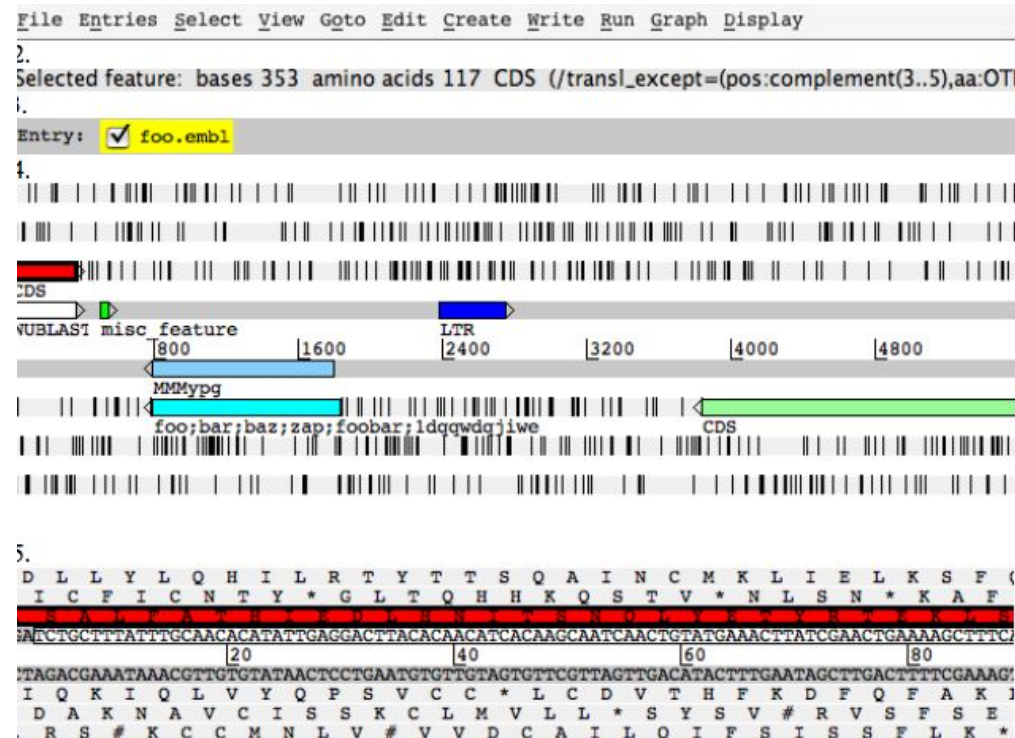
Visualization

Visualisation

- Genome annotation visualisation (`Artemis`, `Icarus`)
- Visualise pan-genome analysis (`Phandango`, `PanExplorer`, `PanACEA`)
- Integrated analysis (`MEGAN`)
- Phylogenetic tree visualisation and annotation (`iTOL`, `R packages`)
- QC visualisation (`multiQC`)
- Assembly quality visualisations (`QUAST/Icarus`, `R/ggplot`)

Genome annotation visualization

Artemis - a **free genome browser and annotation tool** that allows visualisation of sequence features, next generation data and the results of analyses within the context of the sequence, and also its six-frame translation.



Genome annotation visualization

Icarus - a novel genome visualizer for accurate **assessment** and **analysis** of **genomic draft assemblies**, which is based on **QUAST** genome quality assessment tool.

- Contig **alignment** viewer - Assembly alignments as well as genes, operons, and reads coverage distribution along the genome.
- Contig **size** viewer – highlights N50, N75 (NG50, NG75) and long contigs

Thanks for your attention!